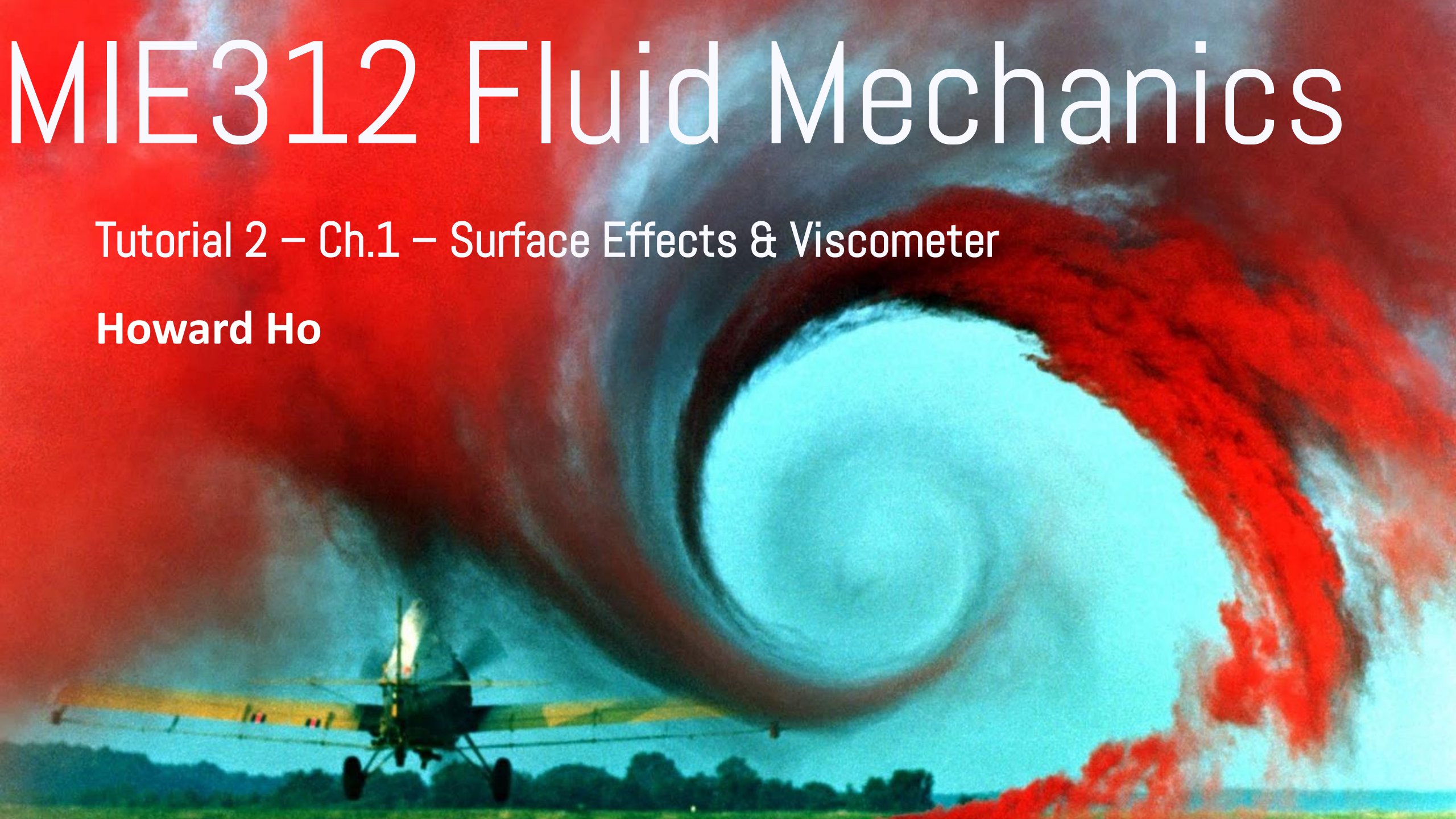
MIE312 Fluid Mechanics

Tutorial 2 – Ch.1 – Surface Effects & Viscometer

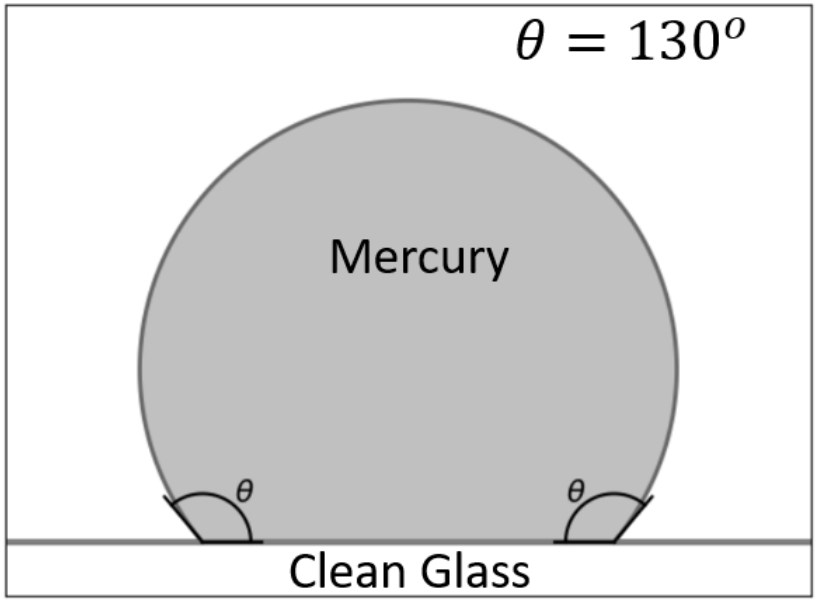
Howard Ho



Learning Objective

- Understand the two surface effects of a liquid
  - Surface tension σ
  - Contact angle θ
- Calculate capillary tubes reading errors
- Understand how viscometers measure viscosity





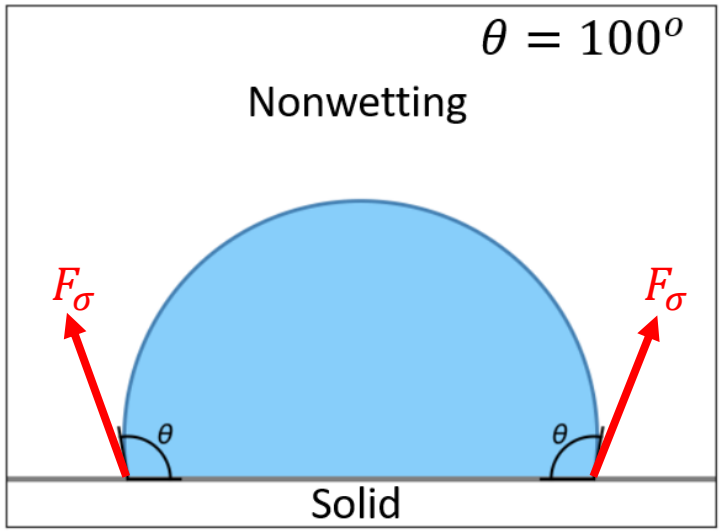


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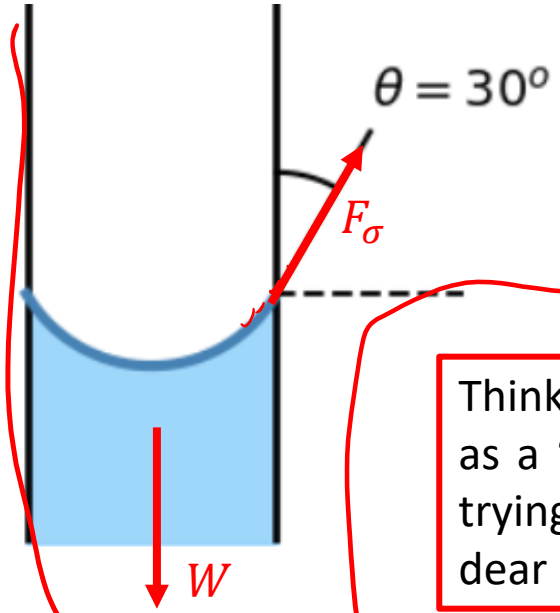
Surface Tension σ



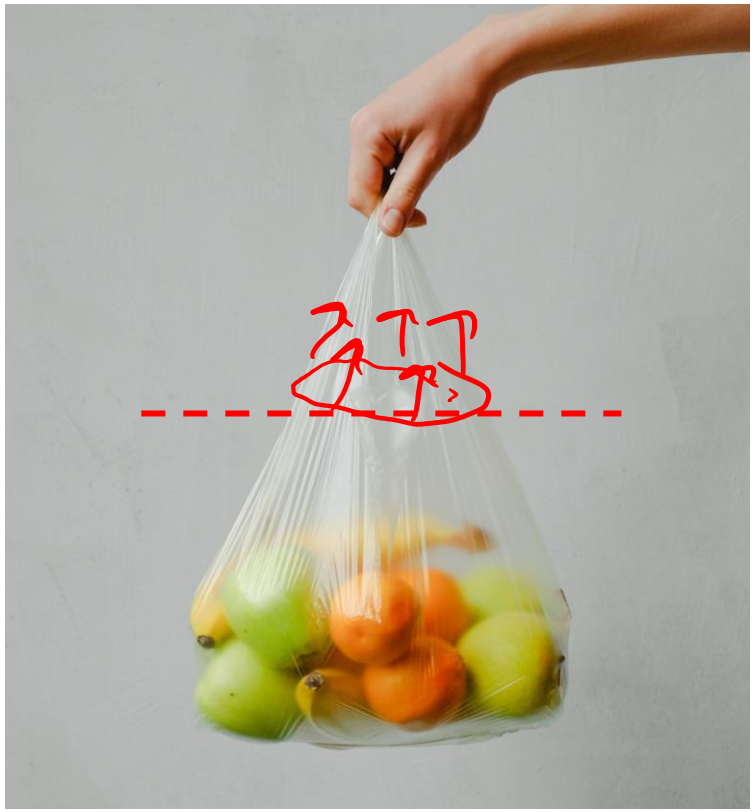
- Not to be confused with Buoyancy (floating is density based)
- Coefficient of surface tension Υ ("upsilon")
  - Unit: N/m or lbf/ft
- Cut length dL
- Surface tension force: $F_\sigma = dL \times \Upsilon$
- The force acts tangent to the edge of the liquid



σ ("sigma")



Think of surface tension as a "cheap" plastic bag, trying to hold onto its dear life… 🥲

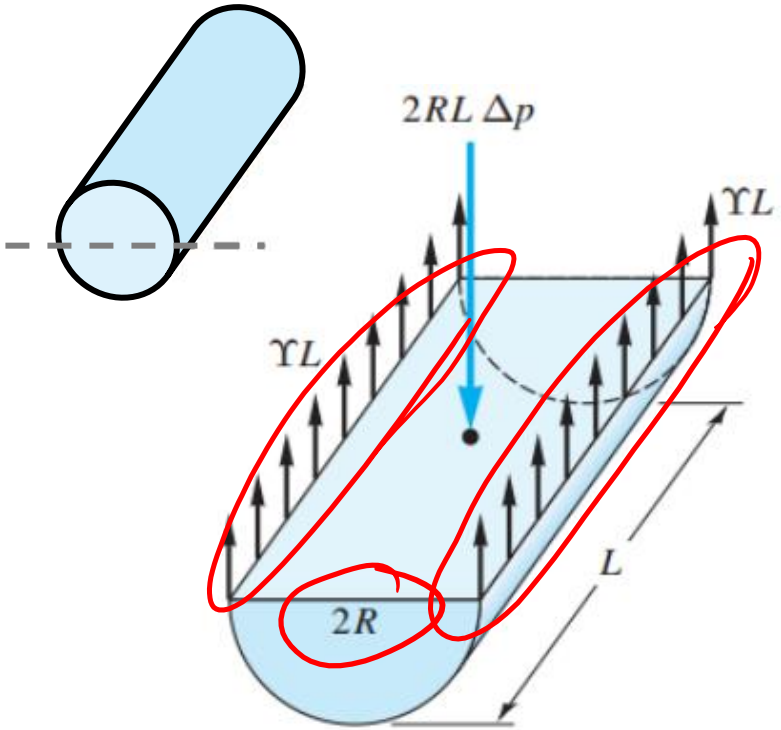


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Cut Length Examples

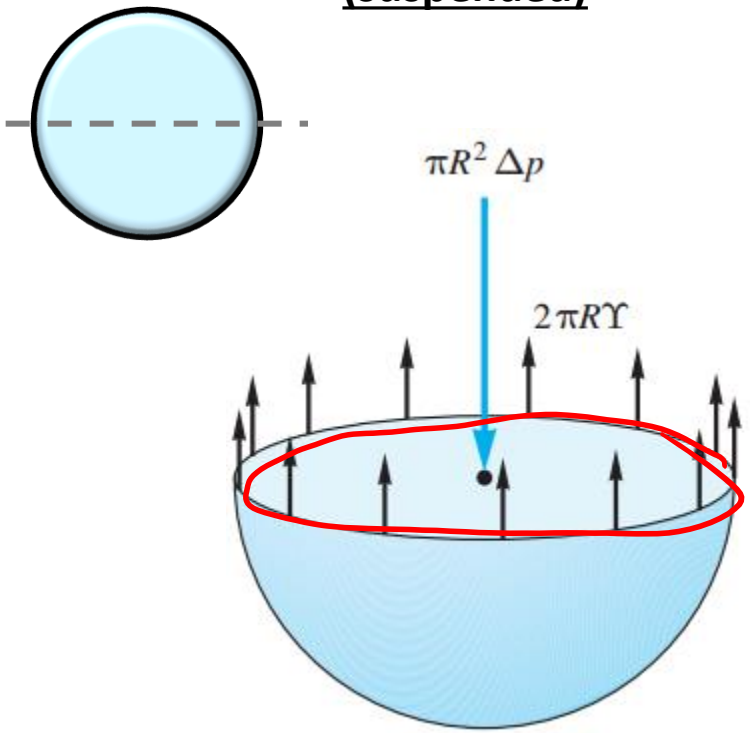
$F_\sigma = dL \times \Upsilon$
dL = cut length

(a) Horizontal Water Pipe



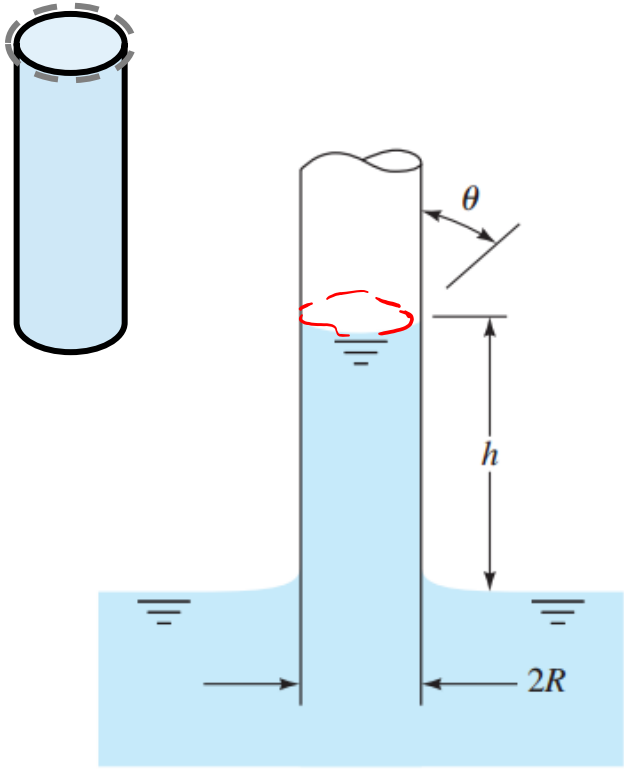
$dL = 2\,L$

(b) Water Droplet (suspended)



$dL = 2\pi R$

(c) Vertical Water Pipe



$dL = 2\pi R$

For geometry (a), we ignore the ends of the cylinder because the cylinder has an "infinite" length ($R \ll L$)

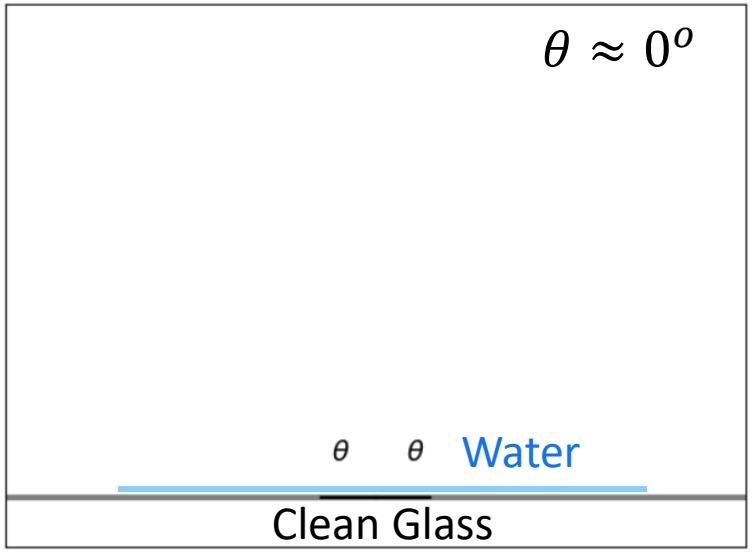
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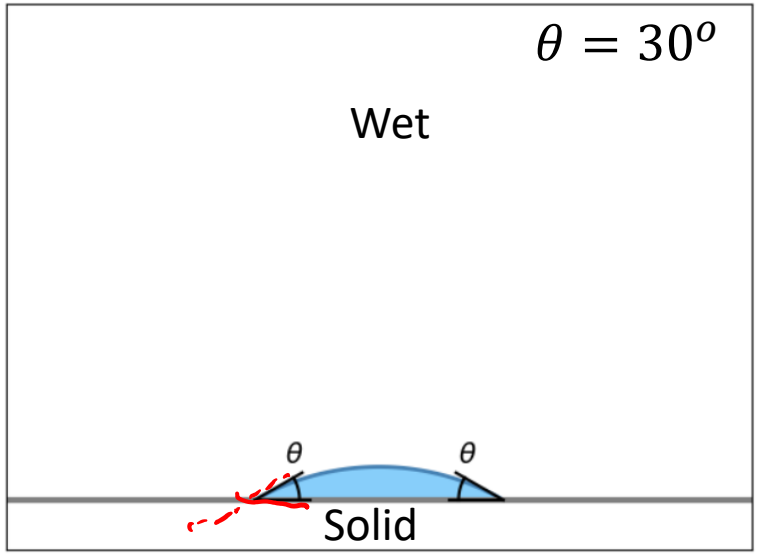
Contact Angle θ

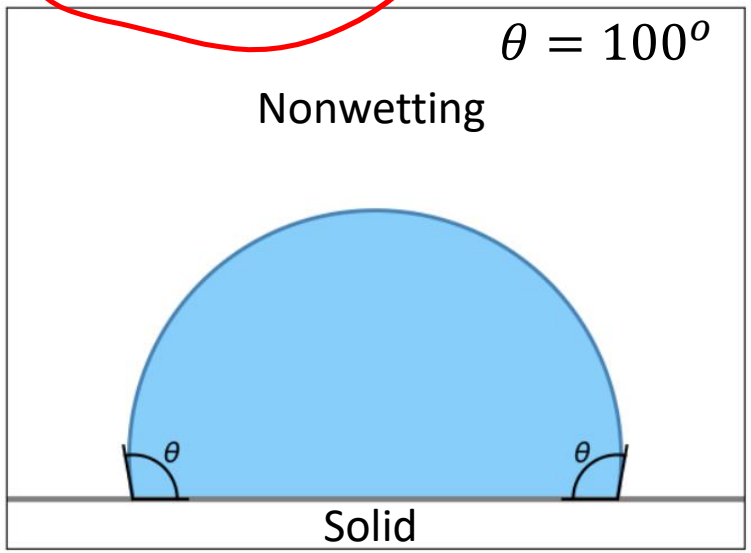
- Contact angle θ is used to describe if the liquid is "wetting" a surface
  - Wet: $\theta < 90^o$
  - Nonwetting: $\theta > 90^o$
- Surface tension coefficient Υ

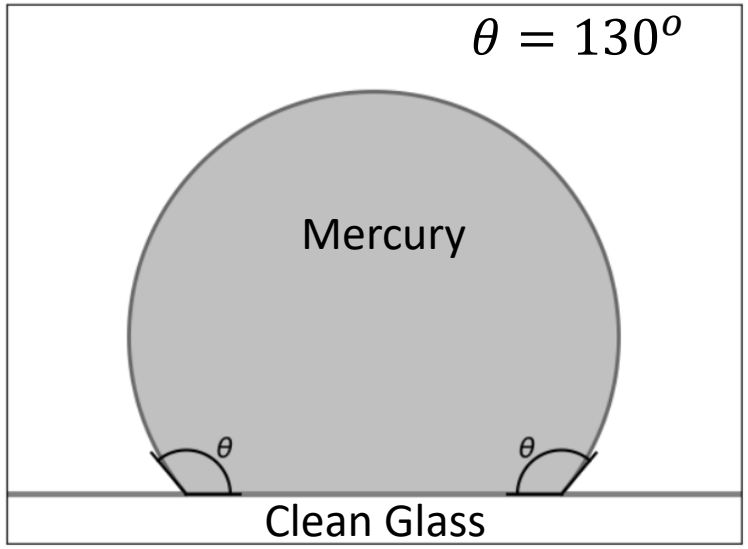
Between medias

$$Y = \begin{cases} 0.0050 \text{ lbf/ft} = 0.073 \text{ N/m} & \text{air–water} \\ 0.033 \text{ lbf/ft} = 0.48 \text{ N/m} & \text{air–mercury} \end{cases} \quad (1.29)$$



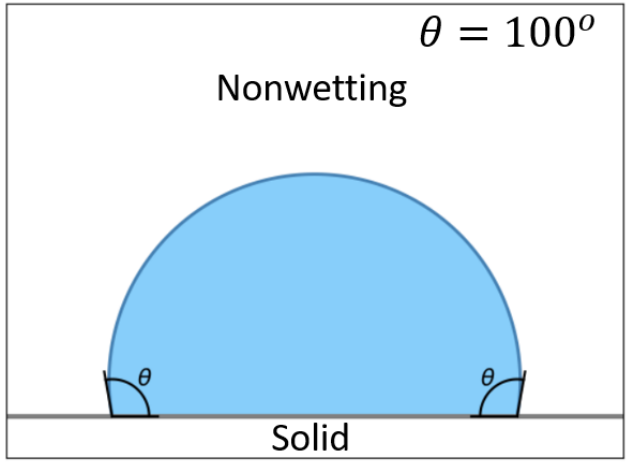






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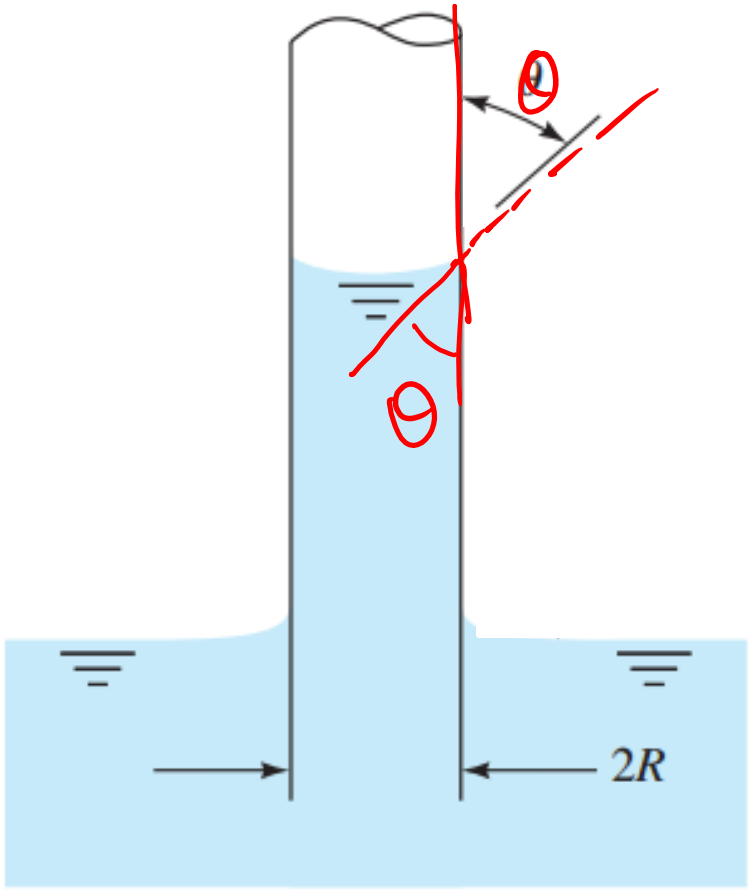
How about vertical tubes?*



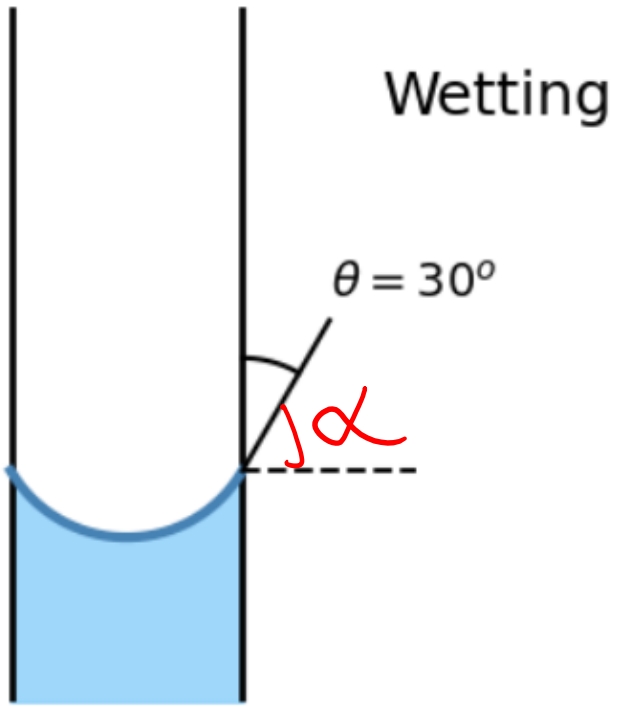
Wet: $\theta < 90^o$

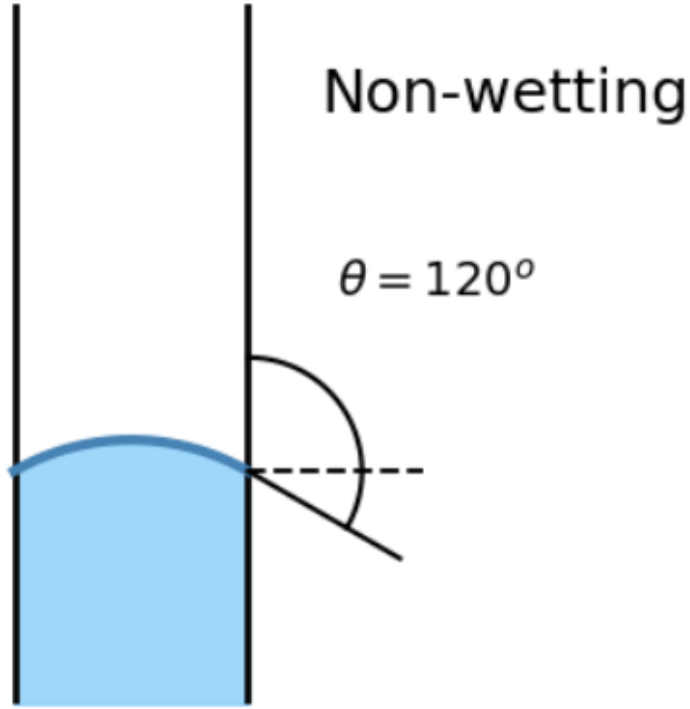
Nonwetting: $\theta > 90^o$

Capillary tubes



Imagine drawing a line from the edge of water through the tube

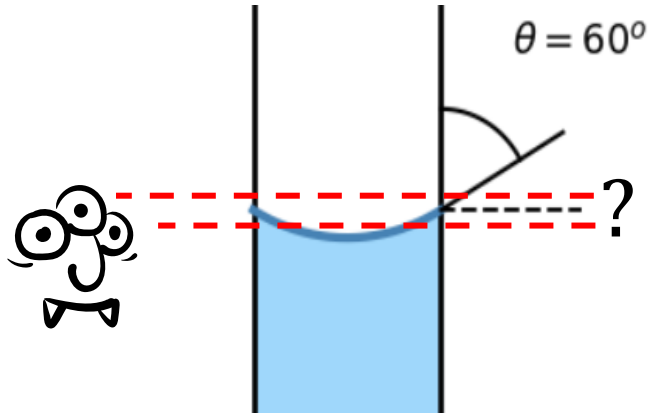


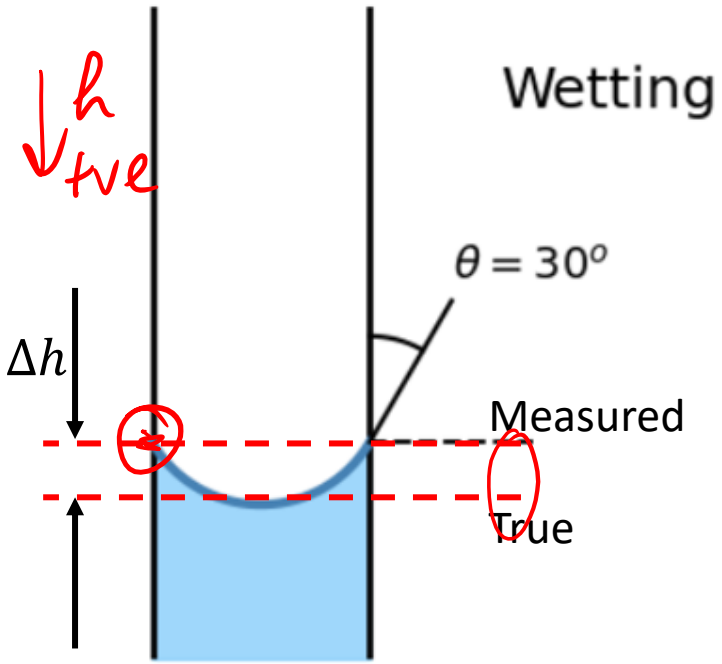


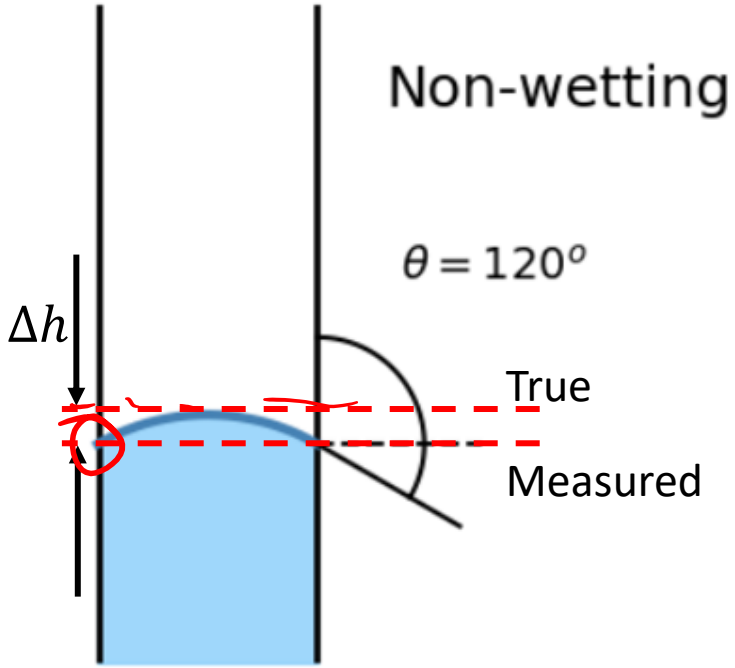
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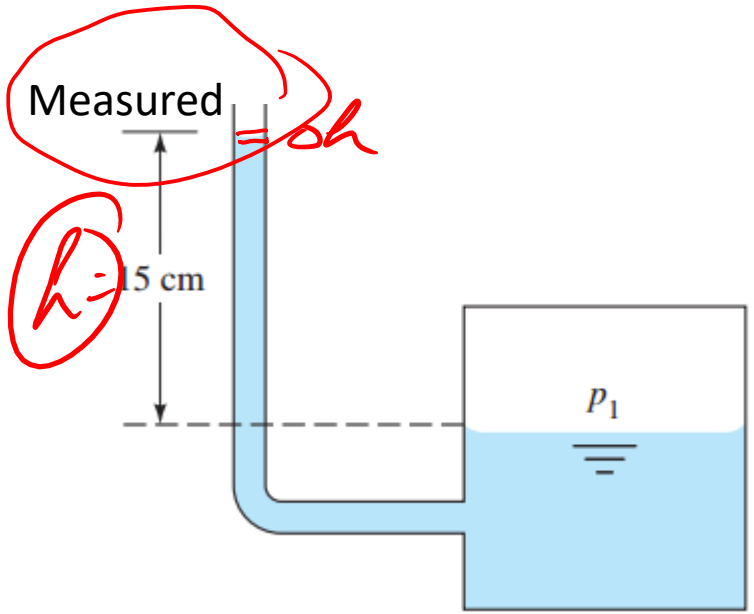
Capillary Tube Error

- Capillary rise/depression relative to liquid surface:
  - $\Delta h = \frac{2\gamma \cos\theta}{\rho g R}$ (I recommend using Δh instead of h to avoid confusion)
- To calculate error from surface tension:
  - $\%\ error = \frac{Measured - True}{True} \times 100\%$









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Question Time



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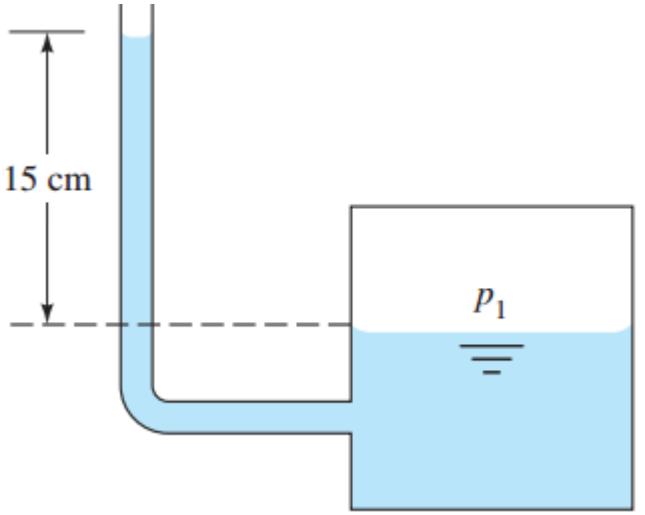
Problem 1

The system in Fig. P1.65 is used to estimate the pressure p_1 in the tank by measuring the 15-cm height of liquid in the 1-mm-diameter tube. Calculate:

(a) The true fluid height in the tube;

(b) The percent error due to capillarity.

$\Upsilon = 0.47\,N/m, \theta = 130^o, \gamma = 132200\ N/m^3, R = 0.5mm$



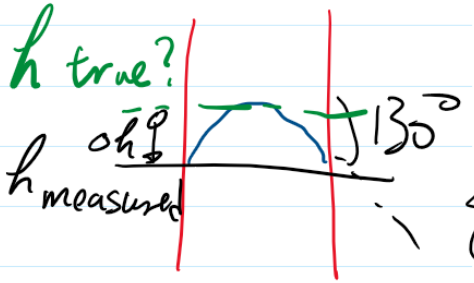
$\Upsilon = 0.47\,N/m, \theta = 130^o, \gamma = 132200\ N/m^3, R = 0.5mm$

gamma

$\gamma^e \neq \gamma = \rho g$

(a)

$\Delta h = \frac{2\gamma^e \cos\theta}{R\,\rho g}$

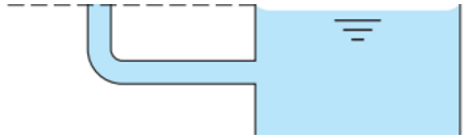


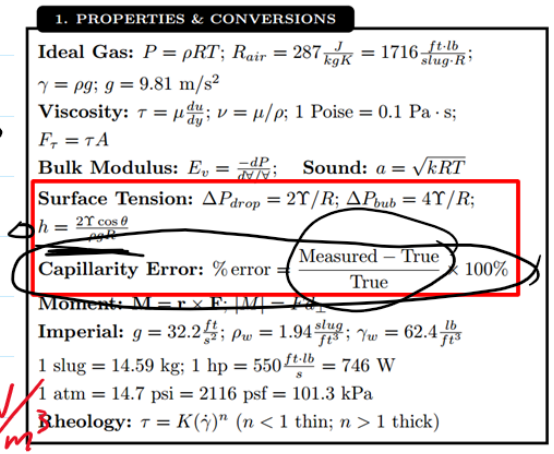
$\theta = 130°,\ \gamma^e = 0.47,\ R = 0.5mm,\ \gamma = \rho g = 132200\ N/m^3$

$\Delta h = \frac{2(0.47)\cos 130°}{(0.5\times10^{-3})(132200)}$

$\Delta h = -9.14 \times 10^{-3}\ m$

Flip sign of Δh because $\theta > 90°$





1. PROPERTIES & CONVERSIONS

Ideal Gas: $P = \rho RT$; $R_{air} = 287\frac{J}{kgK} = 1716\frac{ft\cdot lb}{slug\cdot R}$;
$\gamma = \rho g$; $g = 9.81\ \text{m/s}^2$
Viscosity: $\tau = \mu\frac{du}{dy}$; $\nu = \mu/\rho$; 1 Poise = 0.1 Pa · s;
$F_\tau = \tau A$
Bulk Modulus: $E_v = \frac{-dP}{dV/V}$; **Sound:** $a = \sqrt{kRT}$
Surface Tension: $\Delta P_{drop} = 2\Upsilon/R$; $\Delta P_{bub} = 4\Upsilon/R$;
$h = \frac{2\Upsilon\cos\theta}{\rho g R}$
Capillarity Error: $\%\,\text{error} = \frac{\text{Measured} - \text{True}}{\text{True}} \times 100\%$
Moment: $\mathbf{M} = \mathbf{r}\times\mathbf{F}$; $|M| = Fd_\perp$
Imperial: $g = 32.2\frac{ft}{s^2}$; $\rho_w = 1.94\frac{slug}{ft^3}$; $\gamma_w = 62.4\frac{lb}{ft^3}$
1 slug = 14.59 kg; 1 hp = $550\frac{ft\cdot lb}{s}$ = 746 W
1 atm = 14.7 psi = 2116 psf = 101.3 kPa
Rheology: $\tau = K(\dot\gamma)^n$ ($n < 1$ thin; $n > 1$ thick)

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Rheology: $\tau = K(\dot\gamma)^n$ ($n < 1$ thin; $n > 1$ thick)

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$h_{true} = 0.15 + 9.14 \times 10^{-3}$

$h_{measured} \quad \delta h$

$= 0.15914\,m$

or $15.91\,cm$ //

b) % Error: $\dfrac{h_{measured} - h_{True}}{h_{True}} \times 100\%$

$0.15 \quad 0.15914 \quad 0.15914$

$= -5.7\%$ //

Surface Tension: $\Delta P_{drop} = 2\Upsilon/R$; $\Delta P_{bub} = 4\Upsilon/R$;
$h = \frac{2\Upsilon\cos\theta}{\rho g R}$

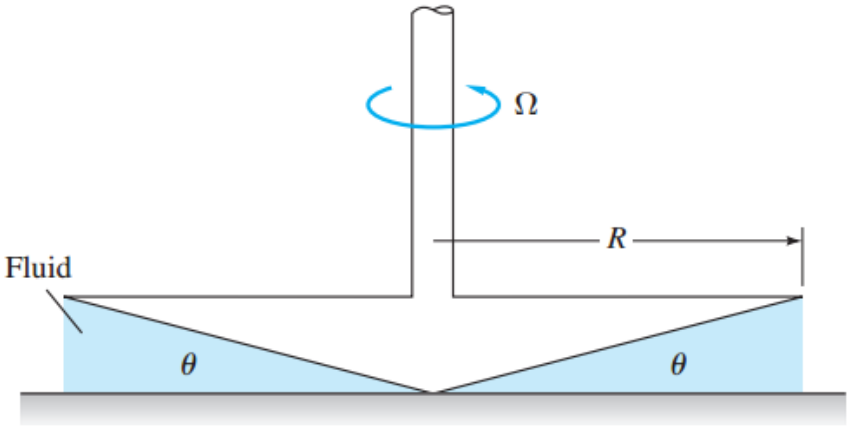
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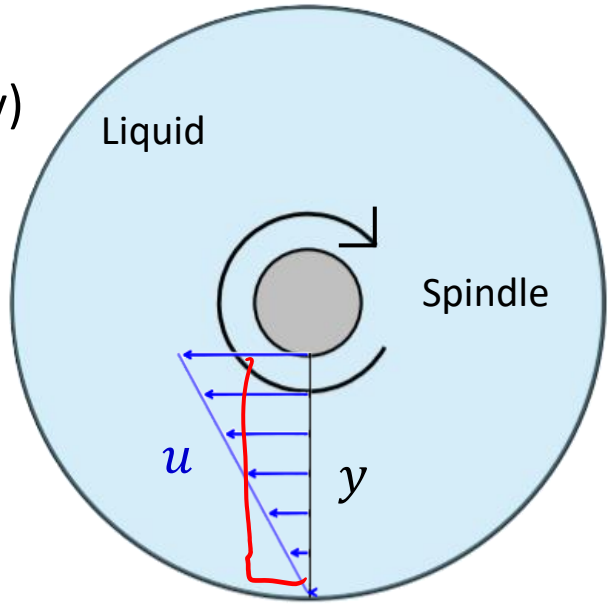
Rotating Viscometer

- Used to measure viscosity of fluids
- Shear stress τ is generated from the rotating spindle
- Recall $\tau = \mu \frac{du}{dy}$
- Torque from the shear stress is measure by the motor
- More torque = More viscous
- They come in different shapes

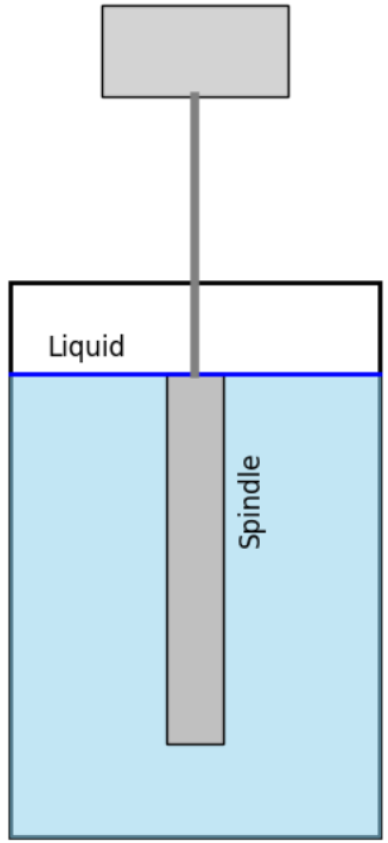
(The cone one works a little differently)



Top-down View



Side View







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Problem 2

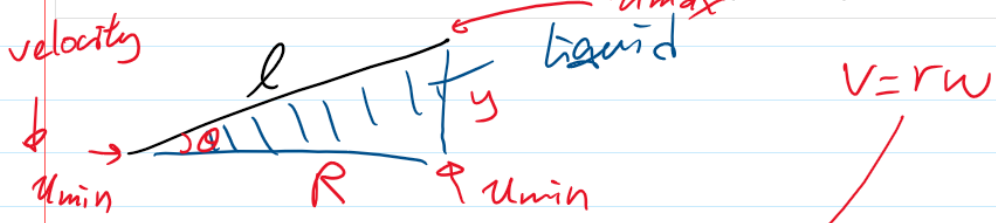
The image display a cone-plate viscometer. The angle of the cone is very small, so that sin θ < θ , and the gap is filled with the test liquid. The torque M to rotate the cone at a rate ω is measured. Assuming a linear velocity profile in the fluid film, derive an expression for the torque M measured as a function of (μ, R, ω, θ).

Hint:

$$dA = \frac{2\pi r}{\cos\theta} dr$$

$$dM = r\, dF_{shear}$$

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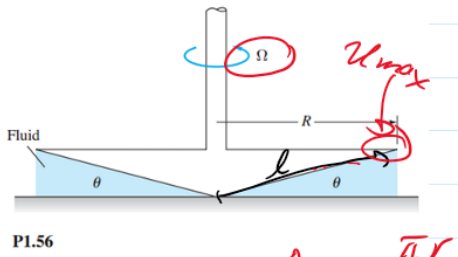
$\tan\theta = \frac{y}{R}$

$y = R\tan\theta$

$u_{max} = R\omega$

$V = r\omega$

Linear velocity profile.



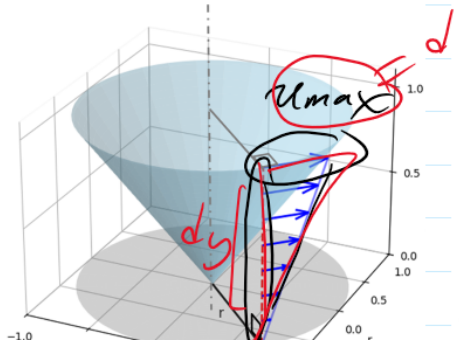
P1.56

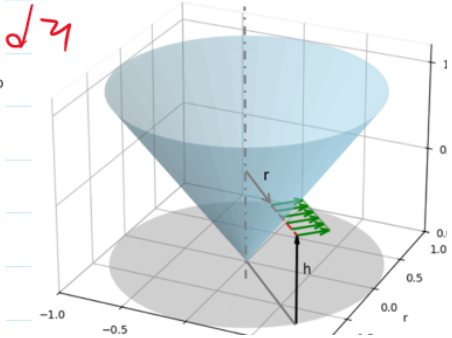
Hint:

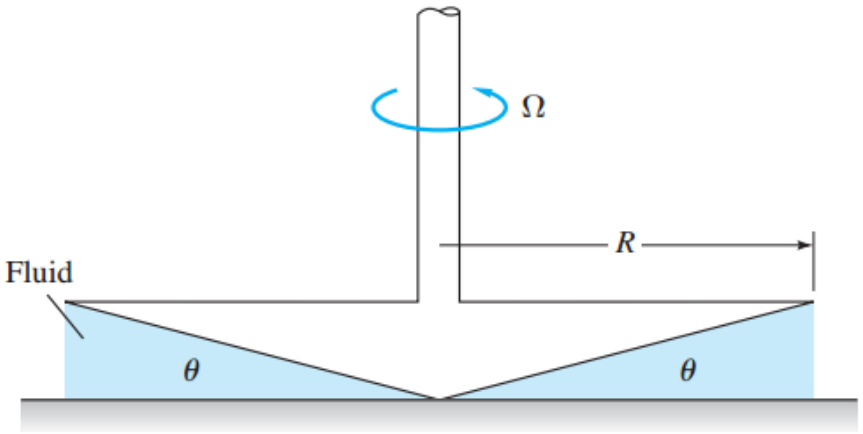
$$dA = \frac{2\pi r}{\cos\theta} dr$$

$$dM = r\, dF_{shear}$$

$A = \frac{\pi r^2}{\cos\theta}$







P1.56

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Linear velocity profile.

$$\frac{du}{dy} = \frac{Rw}{R\tan\theta}$$

$$\tau = \mu \frac{du}{dy}$$

(shear, viscosity)

$$\tau = \mu \frac{w}{\tan\theta}$$

Must use differential form.

$$dF_{shear} = \tau \, dA$$

Torque (M) is not uniform (more torque as we more up)

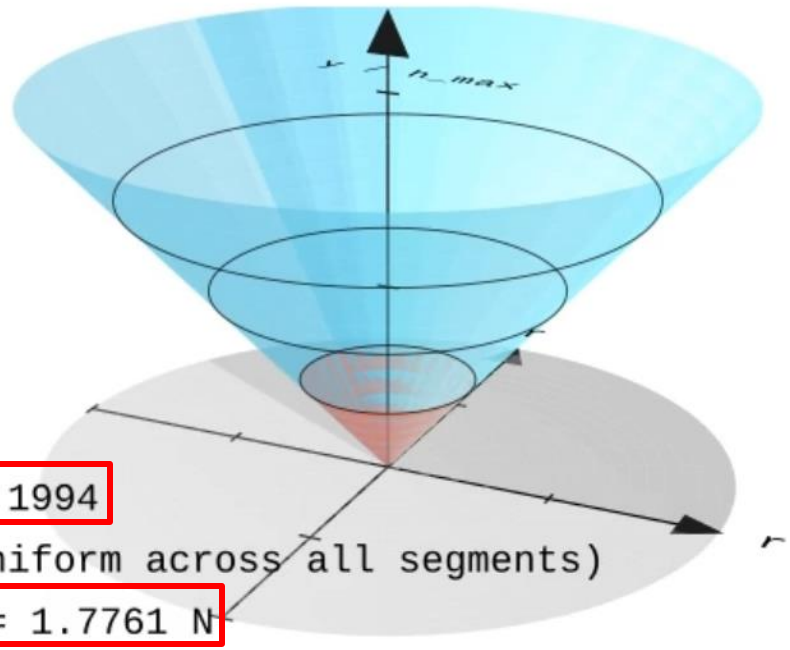
$$dM = r \, dF_{shear}$$

$$\int dM = \int_0^R r \left(\mu \frac{w}{\tan\theta}\right) \frac{2\pi r}{\cos\theta} dr$$

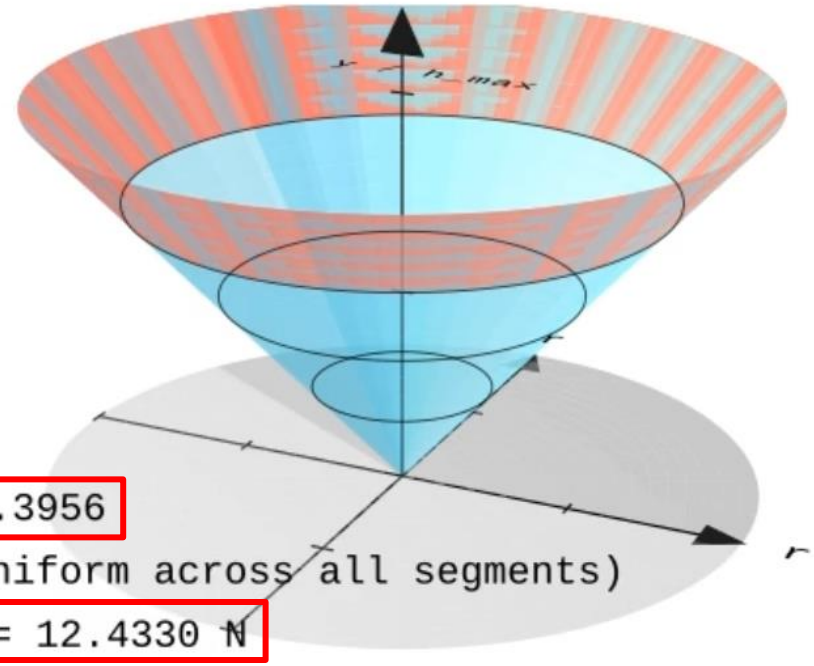
(τ, dA)

$$M = \frac{2\mu\pi w}{\sin\theta} \frac{R^3}{3}$$

The local shear force increases as we move up, Torque $M = r \times F_{shear}$, need integration for this (otherwise we're treating at as a circular cylinder.



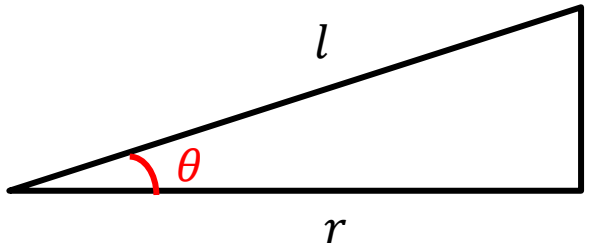
```
Segment 1 / 4
Altitude y/h_max
= 0.00 to 0.25
Contact area A = 0.1994
tau = 8.908 Pa  (uniform across all segments)
F_shear = tau * A = 1.7761 N
```



```
Segment 4 / 4
Altitude y/h_max
= 0.75 to 1.00
Contact area A = 1.3956
tau = 8.908 Pa  (uniform across all segments)
F_shear = tau * A = 12.4330 N
```

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You can also drive the dA equation from scratch, since the top is not in contact with the liquid, we can ignore the flat bit,



$$A_{cone} = \pi r l$$

$$dA = \pi(l\, dr + r\, dl) - (1)$$

$$\cos\theta = \frac{r}{l}$$

$$l = \frac{r}{\cos\theta} - (2)$$

$$dl = \frac{1}{\cos\theta} dr - (3)$$

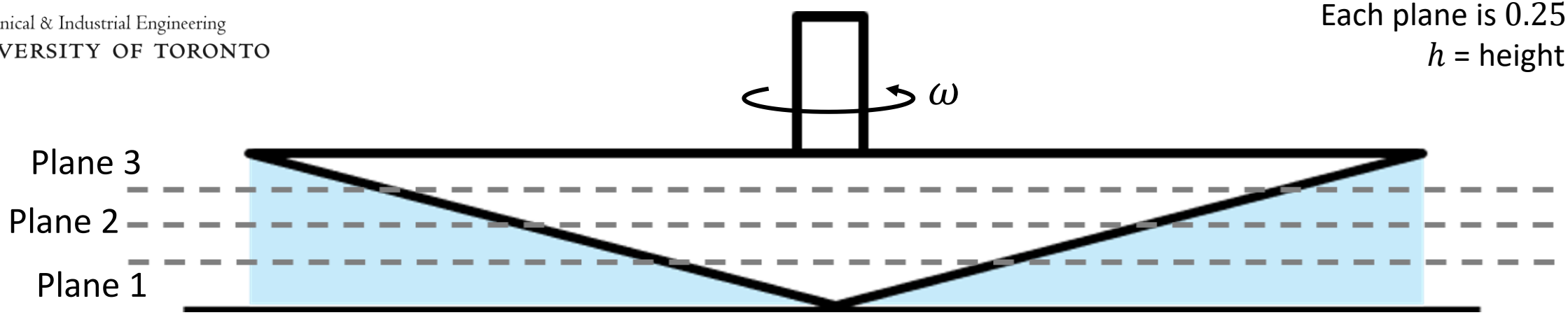
sub. (2) and (3) Into (1),

$$dA = \pi\left(\frac{r}{\cos\theta}\, dr + \frac{r}{\cos\theta} dr\right)$$

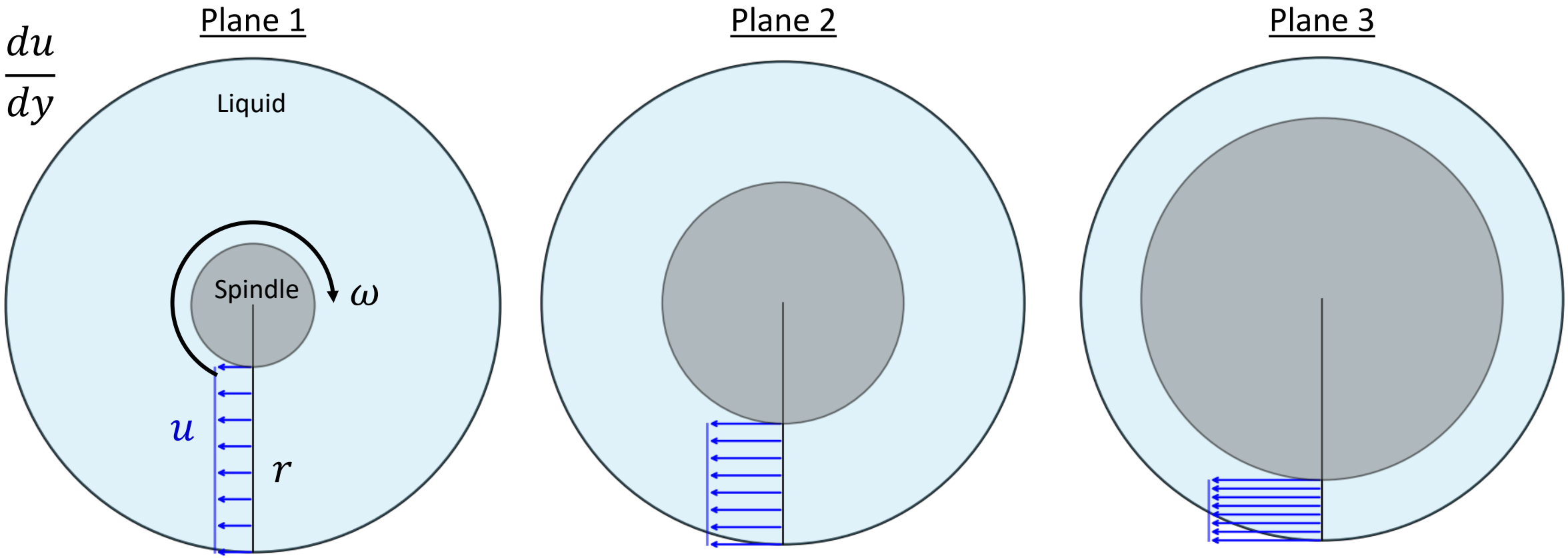
$$dA = \frac{2\pi r}{\cos\theta} dr$$

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Each plane is $0.25\ h$ apart
h = height of cone



$$\tau = \mu \frac{du}{dy}$$

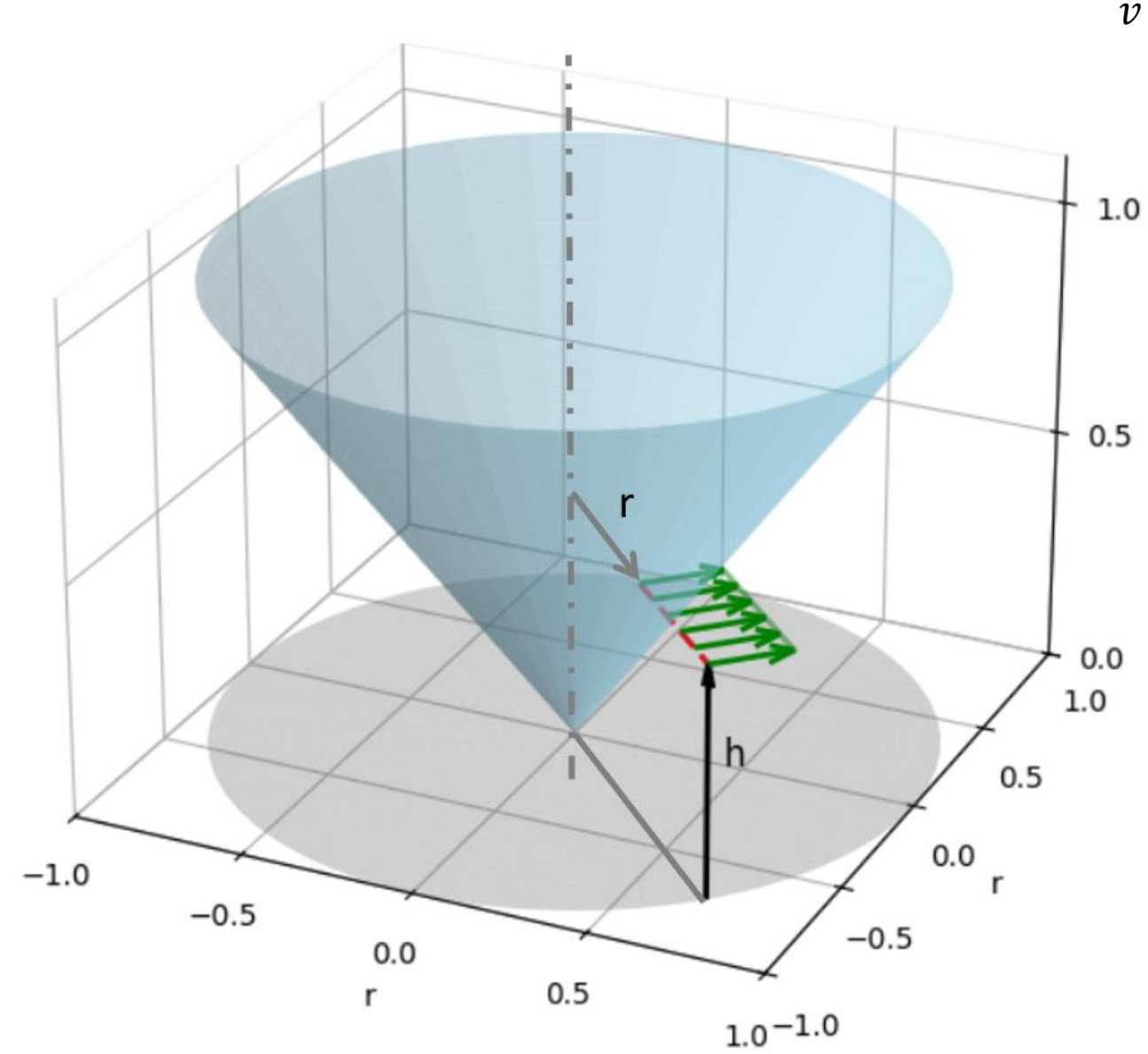


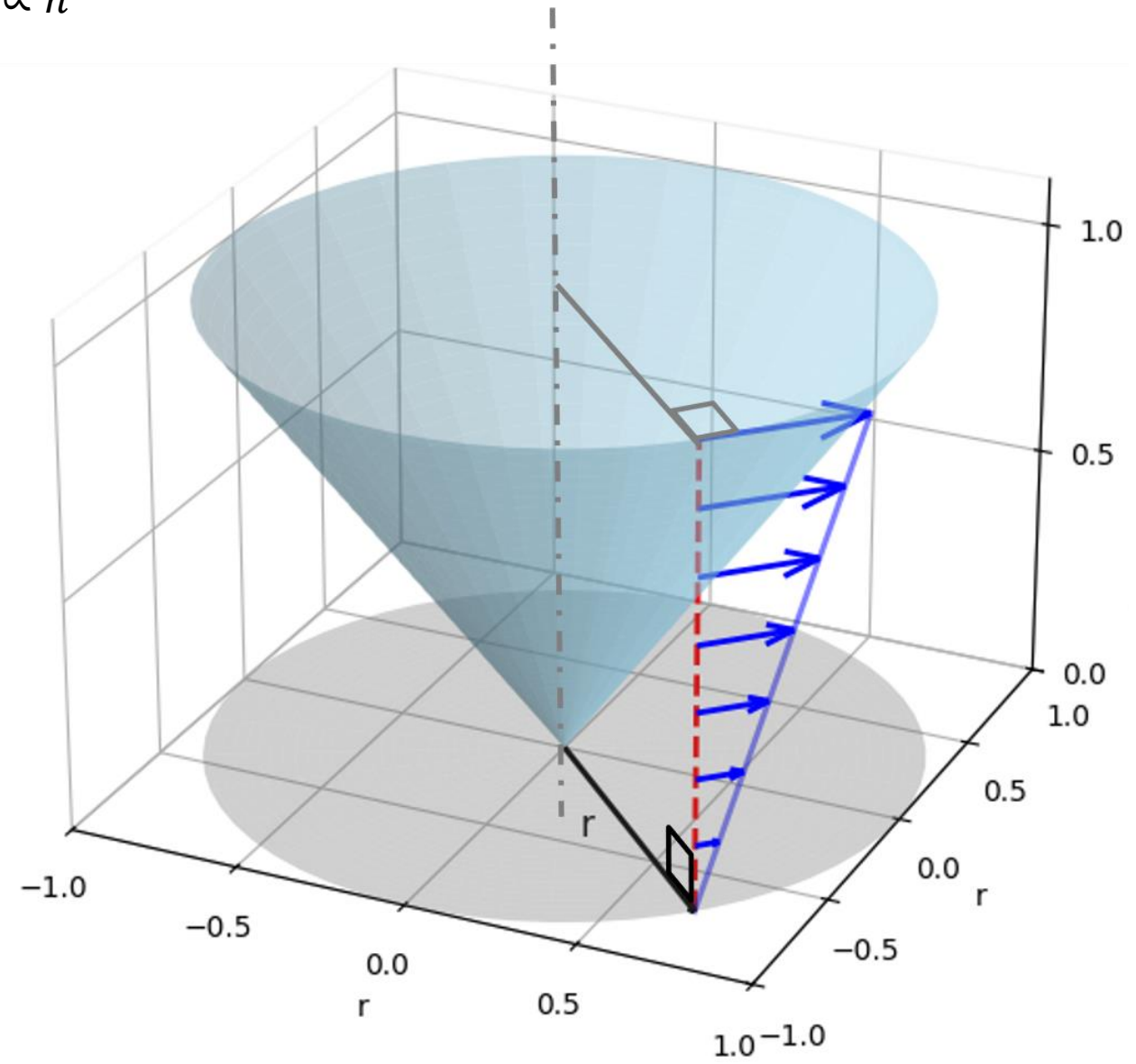
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$$v = r\omega$$

But we are using the cone's r at the local altitude

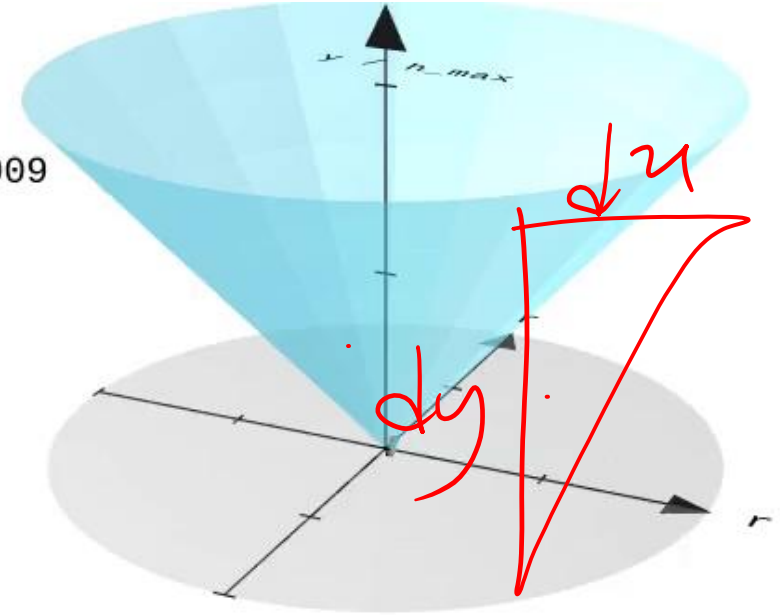
$$v \propto h$$



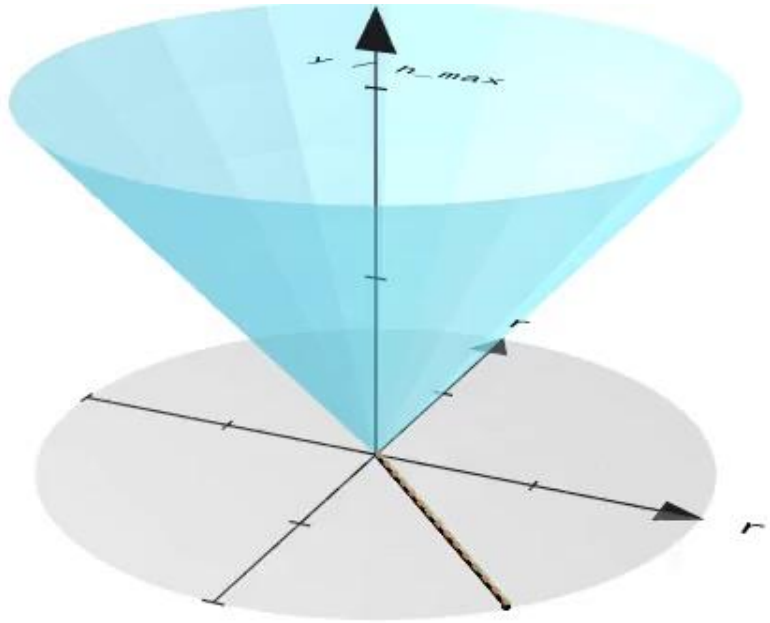


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Cone angle = 10°
r = 0.05
Local gap h = 0.009
v_top = 0.08

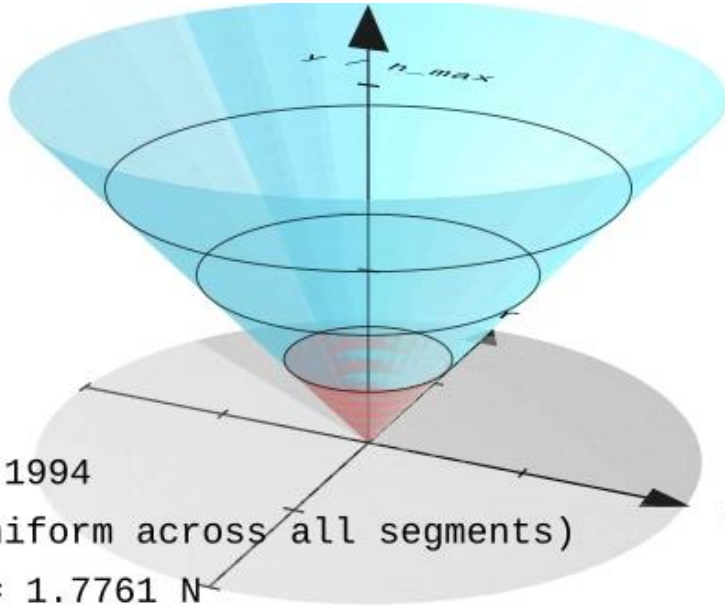


Cone angle = 10°
v_theta = 0.02



Link to animations:
https://howardho-cfd.github.io/teaching/fluids/interactive-tools/cone-viscometer/

$r \times F_{shear}$

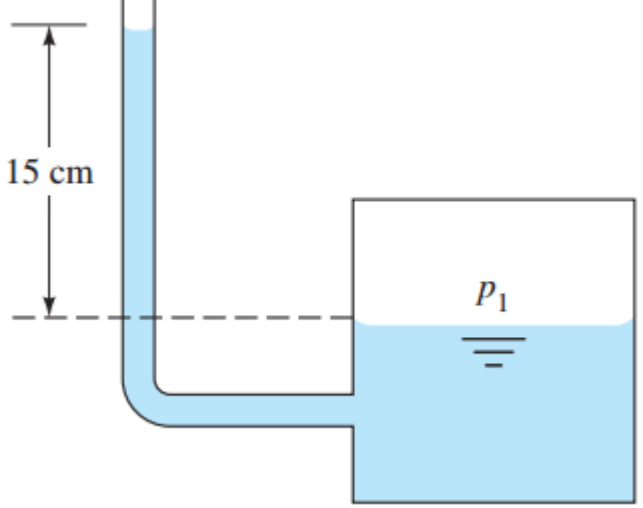


Segment 1 / 4
Altitude y/h_max
= 0.00 to 0.25
Contact area A = 0.1994
tau = 8.908 Pa (uniform across all segments)
F_shear = tau * A = 1.7761 N

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Problem 1

The system in Fig. P1.65 is used to estimate the pressure p_1 in the tank by measuring the 15-cm height of liquid in the 1-mm-diameter tube. The fluid is at 60°C. Calculate the true fluid height in the tube and the percent error due to capillarity if the fluid is (a) water; and (b) mercury.



Solution: This is a somewhat more realistic variation of Ex. 1.9. Use values from that example for contact angle θ:

(a) Water at 60°C: $\gamma \approx 9640\ \text{N/m}^3$, $\theta \approx 0°$:

$$h = \frac{4Y\cos\theta}{\gamma D} = \frac{4(0.0662\ \text{N/m})\cos(0°)}{(9640\ \text{N/m}^3)(0.001\ \text{m})} = 0.0275\ \text{m},$$

or: $\Delta h_{true} = 15.0 - 2.75\ \text{cm} \approx \mathbf{12.25\ cm\ (+22\%\ error)}$ *Ans.* (a)

Δh

"Measured" h

"True" h

Equation for error:

$$\%\ error = \frac{Measured\ -\ True}{True} \times 100\%$$

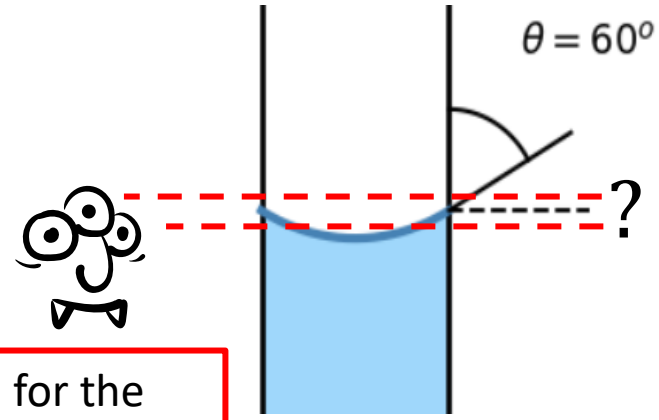
Part (a):

$$\%\ error = \frac{15-12.25}{12.25} \times 100\% = 22.4\%$$

(b) Mercury at 60°C: $\gamma \approx 132200\ \text{N/m}^3$, $\theta \approx 130°$:

$$h = \frac{4Y\cos\theta}{\gamma D} = \frac{4(0.47\ \text{N/m})\cos 130°}{(132200\ \text{N/m}^3)(0.001\ \text{m})} = -0.0091\ \text{m},$$

or: $\Delta h_{true} = 15.0 + 0.91 \approx \mathbf{15.91\ cm\ (-6\%error)}$ *Ans.* (b)



The only difference here compared with Problem 1 is P_1 is enclosed, and they used diameter D for the tube instead of radius, that's why the h equation starts with 4 instead of 2 ($D = 2R$).

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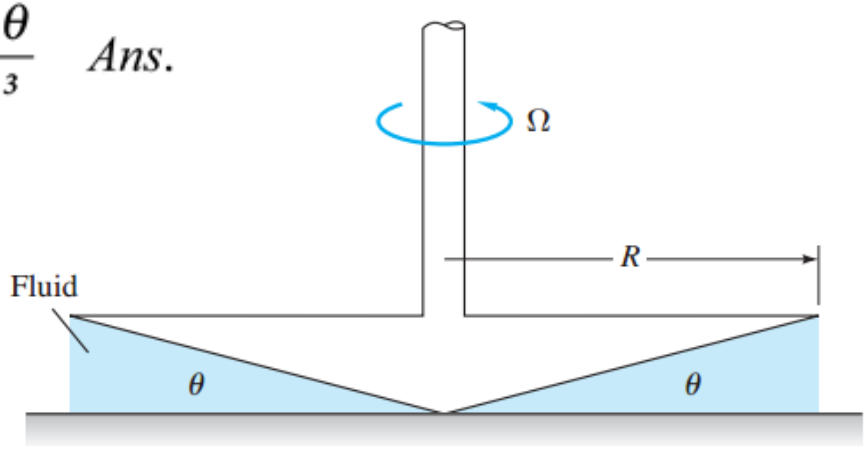
Problem 2

The device in Fig. P1.56 is called a cone-plate viscometer. The angle of the cone is very small, so that sin θ < θ , and the gap is filled with the test liquid. The torque M to rotate the cone at a rate ω is measured. Assuming a linear velocity profile in the fluid film, derive an expression for fluid viscosity μ as a function of (M , R , ω , θ).

Solution: For any radius $r \le R$, the liquid gap is $h = r \tan\theta$. Then

$$d(\text{Torque}) = dM = \tau\, dA_w r = \left(\mu \frac{\Omega r}{r\tan\theta}\right)\left(2\pi r \frac{dr}{\cos\theta}\right) r, \quad \text{or}$$

$$M = \frac{2\pi\Omega\mu}{\sin\theta}\int_0^R r^2\, dr = \frac{2\pi\Omega\mu R^3}{3\sin\theta}, \quad \text{or:} \quad \mu = \frac{3M\sin\theta}{2\pi\Omega R^3} \quad \textit{Ans.}$$



P1.56

Ω is same as ω here, we tend to use lower case ω in fluids

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Link to a pretty cool video

https://www.youtube.com/watch?v=E2unnSK7WTE



This Is Why Water Striders Make Terrible Lifeguards | Deep Look

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